

DEPARTMENT OF MATHEMATICS
MAT111 INTRODUCTORY MATHEMATICS I
WEEK 2: LECTURE NOTE

COMBINATIONS OF FUNCTIONS

Definition 1 (Arithmetic Combinations of Functions). *Let f and g be two functions with overlapping domains. Then, for all x common to both domains, the sum, difference, product, and quotient of f and g are defined as follows:*

1. **Sum:**

$$(f + g)(x) = f(x) + g(x).$$

2. **Difference:**

$$(f - g)(x) = f(x) - g(x).$$

3. **Product:**

$$(fg)(x) = f(x)g(x).$$

4. **Quotient:**

$$\left(\frac{f}{g}\right)(x) = \frac{f(x)}{g(x)}, \quad g(x) \neq 0.$$

Example 2. Let $f(x) = 2x - 3$ and $g(x) = x^2 - 1$. Then

1. $(f + g)(x) = f(x) + g(x) = (2x - 3) + (x^2 - 1) = x^2 + 2x - 4$

2. $(f - g)(x) = f(x) - g(x) = (2x - 3) - (x^2 - 1) = -x^2 + 2x - 2.$

3. $(fg)(x) = f(x)g(x) = (2x - 3)(x^2 - 1) = 2x^3 - 3x^2 - 2x + 3.$

4. $\left(\frac{f}{g}\right)(x) = \frac{f(x)}{g(x)} = \frac{2x - 3}{x^2 - 1}, \quad x \neq \pm 1.$

Example 3. Given $f(x) = 2x + 1$ and $g(x) = x^2 + 2x - 1$, find $(f + g)(x)$. Then evaluate the sum when $x = 2$.

Solution.

(i) $(f + g)(x) = f(x) + g(x) = (2x + 1) + (x^2 + 2x - 1) = x^2 + 4x$

(ii) $(f + g)(2) = 2^2 + 4(2) = 4 + 8 = 12$

Example 4. Arithmetic Combinations of Functions

1. **Finding the Difference of Two Functions:**

Given $f(x) = 2x + 1$ and $g(x) = x^2 + 2x - 1$, find $(f - g)(x)$. Then evaluate the difference when $x = 2$.

(i) $(f - g)(x) = f(x) - g(x) = (2x + 1) - (x^2 + 2x - 1) = -x^2 + 2$

(ii) $(f - g)(2) = -(2)^2 + 2 = -2.$

2. **Finding the Product of Two Functions:**

Given $f(x) = x^2$ and $g(x) = x - 3$, find $fg(x)$. Then evaluate the product when $x = 4$.

$$(i) \quad (fg)(x) = f(x)g(x) = x^2(x - 3) = x^3 - 3x^2$$

$$(ii) \quad (fg)(4) = (4)^3 - 3(4)^2 = 64 - 48 = 16$$

3. **Finding the Quotient of Two Functions:**

Given $f(x) = \sqrt{x}$ and $g(x) = \sqrt{4 - x^2}$, find $\left(\frac{f}{g}\right)(x)$. Then find the domain of $\frac{f}{g}$.

Solution. We have

$$\left(\frac{f}{g}\right)(x) = \frac{f(x)}{g(x)} = \frac{\sqrt{x}}{\sqrt{4 - x^2}}.$$

The domain of f is $[0, \infty)$ and the domain of g is $[-2, 2]$. The intersection of these domains is

$$[0, 2].$$

Because $g(2) = 0$, the domain of $\frac{f}{g}$ is

$$[0, 2).$$

COMPOSITION OF FUNCTIONS

Definition 5 (Composition of Two Functions). The composition of two function f and g , denoted by $f \circ g$, is defined as

$$(f \circ g)(x) = f(g(x)).$$

The domain of $f \circ g$ is the set of all x in the domain of g such that $g(x)$ is in the domain of f .

Example 6. (Forming the Composition of f with g):

Find $(f \circ g)(x)$ for $f(x) = \sqrt{x}$, $x \geq 0$ and $g(x) = x - 1$, $x \geq 1$. If possible, find $(f \circ g)(2)$ and $(f \circ g)(0)$.

Solution. We have

$$(f \circ g)(x) = f(g(x)) = f(x - 1) = \sqrt{x - 1}.$$

Therefore,

1. The domain of $f \circ g$ is $[1, \infty)$.

2. $(f \circ g)(0)$ is **not** defined because 0 is **not** in the domain of $f \circ g$

3. $(f \circ g)(2) = \sqrt{2 - 1} = 1$.

Example 7. Given $f(x) = x + 2$ and $g(x) = 4 - x^2$, then

$$(a) \quad (f \circ g)(x) = f(g(x)) = f(4 - x^2) = (4 - x^2) + 2 = -x^2 + 6.$$

$$(b) \quad (g \circ f)(x) = g(f(x)) = g(x + 2) = 4 - (x + 2)^2 = -x^2 - 4x.$$

Example 8. Find the domain of $f \circ g$ for the functions given by $f(x) = x^2 - 9$ and $g(x) = \sqrt{9 - x^2}$.

Solution. We have that

$$(f \circ g)(x) = f(g(x)) = f(\sqrt{9 - x^2}) = (\sqrt{9 - x^2})^2 - 9 = (9 - x^2) - 9 = -x^2.$$

From this, it might appear that the domain of the composition is the set of all real numbers. This, however, is not true. Because the domain of f is $(-\infty, \infty)$ and the domain of g is $[-3, 3]$, the domain of $f \circ g$ is $[-3, 3]$.

Example 9. Given $f(x) = 2x + 3$ and $g(x) = \frac{1}{2}(x - 3)$. Find $f \circ g$ and $g \circ f$.

Solution.

$$\begin{aligned}(f \circ g)(x) &= f(g(x)) = f\left(\frac{1}{2}(x - 3)\right) \\ &= 2 \cdot \frac{1}{2}(x - 3) + 3 \\ &= x\end{aligned}$$

$$\begin{aligned}(g \circ f)(x) &= g(f(x)) = g(2x + 3) \\ &= \frac{1}{2}((2x + 3) - 3) \\ &= x.\end{aligned}$$

INVERSE FUNCTIONS

Definition 10. Let f and g be two functions such that

$$\begin{aligned}(f \circ g)(x) &= x \quad \text{for every } x \text{ in the domain of } g; \\ (g \circ f)(x) &= x \quad \text{for every } x \text{ in the domain of } f.\end{aligned}$$

Under these conditions, the function g is called the inverse function of the function f . The function g is denoted by f^{-1} . So,

$$f(f^{-1})(x) = x \quad \text{and} \quad f^{-1}(f(x)) = x.$$

The domain of f must be equal to the range of f^{-1} , and the range of f must be equal to the domain of f^{-1} .

Remark 11. If the function g is the inverse function of the function f , then it must also be true that the function f is the inverse function of the function g . For this reason, you can say that the functions f and g are inverse functions of each other.

Example 12. (Verifying Inverse Functions Algebraically)

Show that the functions $f(x) = 2x^3 - 1$ and $g(x) = \sqrt[3]{\frac{x+1}{2}}$ are inverse of each other.

Solution. We note that the domain of f is $\mathbb{R}$ and the domain of g is also $\mathbb{R}$. Thus for every $x \in \mathbb{R}$ we have

$$f(g(x)) = f\left(\sqrt[3]{\frac{x+1}{2}}\right) = 2 \cdot \left(\sqrt[3]{\frac{x+1}{2}}\right)^3 - 1 = 2 \cdot \frac{x+1}{2} - 1 = x$$

and

$$g(f(x)) = g(2x^3 - 1) = \sqrt[3]{\frac{2x^3 - 1 + 1}{2}} = \sqrt[3]{\frac{2x^3}{2}} = \sqrt[3]{x^3} = x$$

Therefore, f and g are inverse functions of each other.

Theorem 13 (Existence of an Inverse Function). A function f has an inverse f^{-1} if and only if f is one-to-one function.

Example 14 (Testing for One-to-One Functions). Is the function $f(x) = \sqrt{x} + 1$ one-to-one.

Solution. The domain of f is the set of all nonnegative real numbers. Let a and b be nonnegative real numbers. Then

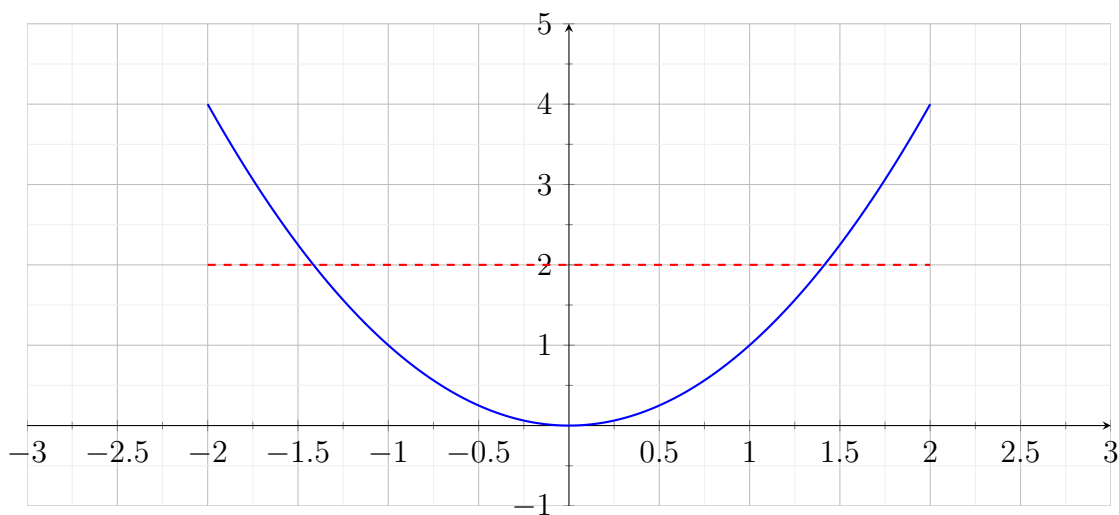
$$\begin{aligned} f(a) = f(b) &\implies \sqrt{a} + 1 = \sqrt{b} + 1 \\ &\implies \sqrt{a} = \sqrt{b} \\ &\implies a = b. \end{aligned}$$

So, $f(a) = f(b)$ implies that $a = b$. We can conclude that f is one-to-one and does has an inverse.

HORIZONTAL LINE TEST

From its graph, it is easy to tell whether a function of x is one-to-one. Simply check to see that every horizontal line intersects the graph of the function at most once. This is called **the Horizontal Line Test**.

For instance, consider the graph of $f(x) = x^2$. On the graph, you can find a horizontal line that intersects the graph twice.



FINDING INVERSE FUNCTIONS ALGEBRAICALLY

To find an inverse of the function f use the following guidelines

1. Use the Horizontal Line Test to decide whether f has an inverse function.
2. In the function for $f(x)$, replace $f(x)$ by y .

3. Interchange the roles of x and y , and solve for y .
4. Replace y by $f^{-1}(x)$ in the new function.
5. Verify that f and f^{-1} are inverse functions of each other by showing that the domain of f is equal to the range of f^{-1} , the range of f is equal to the domain of f^{-1} , and $f(f^{-1}(x)) = x$ and $f^{-1}(f(x)) = x$.

Example 15. FINDING INVERSE FUNCTIONS ALGEBRAICALLY

1. Find the inverse function of

$$f(x) = \frac{5 - 3x}{2}.$$

Solution. f has an inverse because it is one-to-one function.

$f(x) = \frac{5 - 3x}{2}$	Write original function.
$f(x) = \frac{5 - 3x}{2}$	Replace $f(x)$ by y
$x = \frac{5 - 3y}{2}$	Interchange x and y
$2x = 5 - 3y$	Multiply each side by 2.
$2x - 5 = -3y$	Isolate the y-term.
$\frac{2x - 5}{-3} = y$	Solve for y .
$f^{-1}(x) = \frac{5 - 2x}{3}$	Replace y by $f^{-1}(x)$

The domains and ranges of f and f^{-1} consist of all real numbers.

We verify that $f(f^{-1}(x)) = x$ and $f^{-1}(f(x)) = x$.

$$f(f^{-1}(x)) = f\left(\frac{5 - 2x}{3}\right) = \frac{5 - 3\left(\frac{5 - 2x}{3}\right)}{2} = \frac{5 - (5 - 2x)}{2} = \frac{2x}{2} = x$$

$$f^{-1}(f(x)) = f^{-1}\left(\frac{5 - 3x}{2}\right) = \frac{5 - 2\left(\frac{5 - 3x}{2}\right)}{3} = \frac{5 - (5 - 3x)}{3} = \frac{3x}{3} = x.$$

2. Find the inverse of $f(x) = x^3 - 4$.

$$\begin{aligned}
 y &= x^3 - 4 \\
 x &= y^3 - 4 \\
 x + 4 &= y^3 \\
 \sqrt[3]{x + 4} &= y \\
 \therefore f^{-1}(x) &= \sqrt[3]{x + 4}.
 \end{aligned}$$

Domain of $f = \mathbb{R} = \text{Range of } f^{-1}$.
Range of $f = \mathbb{R} = \text{Domain of } f^{-1}$.

3. Find the inverse function of $f(x) = \sqrt{2x - 3}$

Solution. f has an inverse on its domain $[\frac{3}{2}, \infty)$.

$$y = \sqrt{2x - 3}$$

$$x = \sqrt{2y - 3}$$

$$x^2 = 2y - 3$$

$$x^2 + 3 = 2y$$

$$\frac{x^2 + 3}{2} = y$$

$$\therefore f^{-1}(x) = \frac{x^2 + 3}{2}.$$

Domain of $f = [\frac{3}{2}, \infty) = \text{Range of } f^{-1}$.

Range of $f = [0, \infty) = \text{Domain of } f^{-1}$.